BigCodeBench/859 Evaluations

Task Description

Perform SVM classification on iris dataset with:

- Test size 0.33 for train-test split
- Warning if accuracy < 0.9
- Warning action set to 'always'
- Return tuple: (accuracy, warning_msg or None)

Model 1 Evaluation

model1_efficiency

Total steps: 8

- Step 1: Problem understanding - **Essential**
- Step 2: Dataset splitting requirement - **Essential**
- Step 3: SVM classifier initialization - **Essential**
- Step 4: Making predictions - **Essential**
- Step 5: Computing accuracy - **Essential**
- Step 6: Conditional check for warning - **Essential**
- Step 7: Warning mechanism with 'always' filter - **Essential**
- Step 8: Return tuple structure - **Essential**

model1_logic

Step Coherence:

- Step 1: Starting point
- Step 2: Coherent - identifies split requirement
- Step 3: Coherent - next logical step after data preparation
- Step 4: Coherent - follows training
- Step 5: Coherent - evaluation step
- Step 6: Coherent - implements requirement
- Step 7: Coherent - warning implementation details
- Step 8: Coherent - output formatting

Internal Consistency: No contradictions, thorough understanding of warnings module

model1_completeness

Problem Coverage:

- Load iris dataset: Yes
- Split with test_size=0.33: Yes
- Train SVM: Yes
- Calculate accuracy: Yes
- Warning if < 0.9: Yes
- Warning action 'always': Yes
- Return tuple: Yes

Factual Correctness: Correct use of UserWarning category and filterwarnings

Constraint and edge case handling: Comprehensive coverage

Model 2 Evaluation

model2_efficiency

Total steps: 8

- Step 1: Import statements - **Non-essential** (already in starter code)
- Step 2: Load iris dataset - **Essential**
- Step 3: Split dataset - **Essential**
- Step 4: Create and fit SVM - **Essential**
- Step 5: Predict and calculate accuracy - **Essential**
- Step 6: Set warnings filter - **Essential**
- Step 7: Check and create warning - **Essential**
- Step 8: Return tuple - **Essential**

model2_logic

Step Coherence:

- Step 1: Starting point (though redundant)
- Step 2: Coherent - data loading
- Step 3: Coherent - mentions random_state for reproducibility
- Step 4: Coherent - model creation
- Step 5: Coherent - evaluation

- Step 6: Coherent - warning setup
- Step 7: Coherent - conditional logic
- Step 8: Coherent - output

Internal Consistency: No contradictions, mentions reproducibility consideration

model2_completeness

Problem Coverage:

- Load iris dataset: Yes
- Split with test_size=0.33: Yes
- Train SVM: Yes
- Calculate accuracy: Yes
- Warning if < 0.9: Yes
- Warning action 'always': Yes
- Return tuple: Yes

Factual Correctness: All correct

Constraint and edge case handling: Mentions random_state for reproducibility

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Model 3 Evaluation

model3_efficiency

Total steps: 7

- Step 1: Problem overview - **Essential**
- Step 2: Dataset loading and splitting - **Essential**
- Step 3: SVM instantiation - **Essential**
- Step 4: Predictions and accuracy - **Essential**
- Step 5: Warning condition and setup - **Essential**
- Step 6: Warning generation - **Essential**
- Step 7: Return tuple - **Essential**

model3_logic

Step Coherence:

- Step 1: Starting point
- Step 2: Coherent - combines related tasks

- Step 3: Coherent - model setup
- Step 4: Coherent - evaluation
- Step 5: Coherent - warning requirements
- Step 6: Coherent - implementation details
- Step 7: Coherent - output formatting

Internal Consistency: Consistent throughout, emphasizes both warning display and return

model3_completeness

Problem Coverage:

- Load iris dataset: Yes
- Split with test_size=0.33: Yes
- Train SVM: Yes
- Calculate accuracy: Yes
- Warning if < 0.9: Yes
- Warning action 'always': Yes
- Return tuple: Yes

Factual Correctness: Correct understanding of dual requirement (warn + return)

Constraint and edge case handling: Good coverage

Model 4 Evaluation

model4_efficiency

Total steps: 5

- Step 1: Understand requirements - **Essential**
- Step 2: Identify necessary steps - **Essential** (though verbose)
- Step 3: Design solution - **Essential**
- Step 4: Consider edge cases - **Non-essential** (speculation)
- Step 5: Verify requirements - **Non-essential** (meta-discussion)

model4_logic

Step Coherence:

- Step 1: Starting point
- Step 2: Coherent - breaks down tasks

- Step 3: Coherent - implementation approach
- Step 4: Coherent - but speculative
- Step 5: Coherent - but redundant verification

Internal Consistency: No contradictions, suggests random_state=42

model4_completeness

Problem Coverage:

- Load iris dataset: Yes
- Split with test_size=0.33: Yes
- Train SVM: Yes
- Calculate accuracy: Yes
- Warning if < 0.9: Yes
- Warning action 'always': Yes
- Return tuple: Yes

Factual Correctness: Correct, mentions reproducibility benefits

Constraint and edge case handling:

- Discusses potential accuracy variations
- Suggests random_state for consistency

Model 5 Evaluation

model5_efficiency

Total steps: 5

- Step 1: Task overview - **Essential**
- Step 2: Load and split data - **Essential**
- Step 3: Train and predict - **Essential**
- Step 4: Accuracy and warning - **Essential**
- Step 5: Random_state discussion - **Non-essential** (explains omission)

model5_logic

Step Coherence:

- Step 1: Starting point
- Step 2: Coherent - data preparation

- Step 3: Coherent - model operations
- Step 4: Coherent - evaluation and output
- Step 5: Coherent - but discusses what's not required

Internal Consistency: No contradictions, notes stratification benefit

model5_completeness

Problem Coverage:

- Load iris dataset: Yes
- Split with test_size=0.33: Yes
- Train SVM: Yes
- Calculate accuracy: Yes
- Warning if < 0.9: Yes
- Warning action 'always': Not explicitly mentioned
- Return tuple: Yes

Factual Correctness: Correct about stratification, though warning 'always' not explicit

Constraint and edge case handling: Notes stratification ensures balanced splits

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Model 6 Evaluation

model6_efficiency

Total steps: 5

- Step 1: Understanding task - **Essential**
- Step 2: Return requirements - **Essential**
- Step 3: Implementation approach - **Essential**
- Step 4: Warning mechanism - **Essential**
- Step 5: Final check and return - **Essential**

model6_logic

Step Coherence:

- Step 1: Starting point
- Step 2: Coherent - output specification
- Step 3: Coherent - implementation steps
- Step 4: Coherent - warning details

- Step 5: Coherent - conditional logic

Internal Consistency: No contradictions, clear understanding

model6_completeness

Problem Coverage:

- Load iris dataset: Yes
- Split with test_size=0.33: Yes
- Train SVM: Yes
- Calculate accuracy: Yes
- Warning if < 0.9: Yes
- Warning action 'always': Yes
- Return tuple: Yes

Factual Correctness: All correct

Constraint and edge case handling: Basic coverage

Summary

- **Most Complete**: Model 1 (thorough warning implementation details)
- **Most Efficient**: Models 5 and 6 (5 steps, though Model 5 has one non-essential)
- **Best Logic**: Models 1 and 3 (clear progression and understanding)
- **Issues**: Model 4 has 2 non-essential steps; Model 5 doesn't explicitly mention warning 'always'